NORDKYL

Nordkyl A/S Aarhus Denmark

NK-600

Air-cooled screw chiller 600 kW R-513A

Service Manual

For authorised Nordkyl service partners

Document	NK-SM-600-EN
Revision	C
Date	2025-03-05
Serial numbers	NK600-210001 and later
Controller	NordCtrl 7, software 6.9 and later
Original language	English

About this manual

This manual is for service technicians who are trained on Nordkyl NK chillers and hold the refrigerant handling certificate required where the unit is installed. It covers diagnosis, the service menus, work on the refrigerant circuits and planned maintenance of the NK-600. Lifting, foundations and first installation are covered in the Installation Manual NK-IM-600. Day-to-day operation by site staff is covered in the User Guide NK-UG-600. The wiring diagram for each unit is kept in the electrical panel and is specific to the serial number.

Between revisions of this manual, Nordkyl issues Product Service Notices and, where a defined serial range needs a corrective action, field actions. Check the Nordkyl Service Portal for notices against the serial number before starting a repair. A notice that names the serial number takes precedence over this manual.

Conventions. Pressures shown on the NordCtrl display and given in this manual are absolute (bar a); a gauge manifold reads about 1 bar lower. Temperatures are in C, temperature differences in K. Menu paths are written Main > Service > Circuits > Circuit 1 > EXV. Circuit 1 is the circuit nearest the control panel.

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Revision history

Rev	Date	Changes
A	2021-06-14	First edition. Controller software 6.x.
B	2023-01-20	Section 7.4 EXV service menu added (software 7.0). Alarms L-11 to L-14 added. Filter drier core changed to 63-0410.
C	2025-03-05	Section 8 updated to Regulation (EU) 2024/573. Section 6.3 (diagnosing L-03) added. Typical operating values in 6.4 revised. Annual EXV manual test added to the maintenance plan.

1 Safety

1.1 Who may work on the unit

Work that opens, or connects to, a refrigerant circuit may only be carried out by a person certified under Regulation (EU) 2024/573 and Implementing Regulation (EU) 2015/2067, category I, or under the equivalent scheme where the unit is installed (in the United States, EPA Section 608 Type II or Universal). Electrical work in the panel requires a qualified electrician. A technician in training may assist under the direct supervision of a certified technician but may not connect gauges, recover or charge refrigerant on his or her own.

1.2 Hazards

DANGER
Each circuit holds 24 kg of refrigerant under pressure. Never open a joint, loosen a flare nut or remove a valve core on a pressurised circuit. Recover the refrigerant first.

WARNING
The EC condenser fans start automatically when the controller calls for condensing. Isolate the unit at the main switch before reaching into the fan section. Fan and EXV driver terminals stay live for up to 3 minutes after isolation.

- Refrigerant: liquid R-513A on the skin causes frostbite. In a plant room, a large release displaces air. R-513A is non-flammable (A1), but in contact with flames or hot surfaces it decomposes into toxic and corrosive products.
- Hot surfaces: the discharge line and oil separator can exceed 90 C.
- Sharp edges: condenser fins and panel edges. Wear cut-resistant gloves.

1.3 Pressure equipment

The NK-600 is a pressure equipment assembly to the Pressure Equipment Directive 2014/68/EU. Each circuit has a safety valve on the high side set to the high side design pressure. Never isolate a safety valve from the circuit. Replace safety valves after any discharge and in any case every five years.

1.4 Personal protective equipment

Safety glasses and insulated gloves when connecting hoses or handling cylinders. Hearing protection when working beside running compressors. A portable refrigerant detector when working in a plant room.

2 Unit description and technical data

2.1 Design

The NK-600 is an air-cooled liquid chiller with two fully independent refrigerant circuits. Each circuit has its own semi-hermetic twin-screw compressor with stepless slide valve capacity control, oil separator, V-coil condenser section with four EC fans, filter drier, sight glass and electronic expansion valve (EXV). The two circuits share one dual-circuit shell and tube evaporator, so a fault in one circuit does not stop the other: the unit continues at reduced capacity.

The unit is controlled by a NordCtrl 7 controller with a 7-inch touch display in the panel door, a Modbus RTU

and BACnet IP interface for the building management system, and a USB port for software updates and log export.

2.2 Technical data

Item	Value
Nominal cooling capacity	600 kW (chilled water 12/7 C, ambient 35 C)
Power input, compressors and fans	198 kW
EER at nominal conditions	3.03
Capacity control	Stepless, 12.5 - 100 % of unit capacity
Refrigerant	R-513A, safety group A1, GWP 631
Refrigerant charge	48 kg in total: two circuits of 24 kg each
CO2 equivalent	15.1 t per circuit, 30.3 t per unit
Compressors	2 x semi-hermetic twin-screw, POE oil, 16 L per compressor
Condenser fans	8 x EC axial, 910 mm, speed controlled
Evaporator	Dual-circuit shell and tube, direct expansion
Chilled water flow (nominal / min / max)	28.7 / 14.3 / 43.0 l/s
Water side pressure drop at nominal flow	48 kPa
Max. water side pressure	10 bar
Design pressure PS, high side / low side	24 bar / 16 bar
Leaving water temperature range	4 to 15 C (down to -8 C with glycol option)
Ambient operating range	-15 to +46 C (-25 C with low ambient kit)
Power supply	400 V 3~ 50 Hz, or 460 V 3~ 60 Hz (North American units)
Maximum running current	372 A (400 V)
Sound power level	94 dB(A)
Dimensions L x W x H	7,850 x 2,260 x 2,480 mm
Operating weight	5,620 kg

2.3 Access levels

NordCtrl has three access levels. Level 1 (no password) shows operating values and active alarms and allows alarm reset. Level 2 (service) opens the Service menu, including the EXV menu and manual override. The level 2 password is issued to authorised service partners through the Service Portal. Level 3 is reserved for Nordkyl.

3 Refrigerant circuit

3.1 Schematic

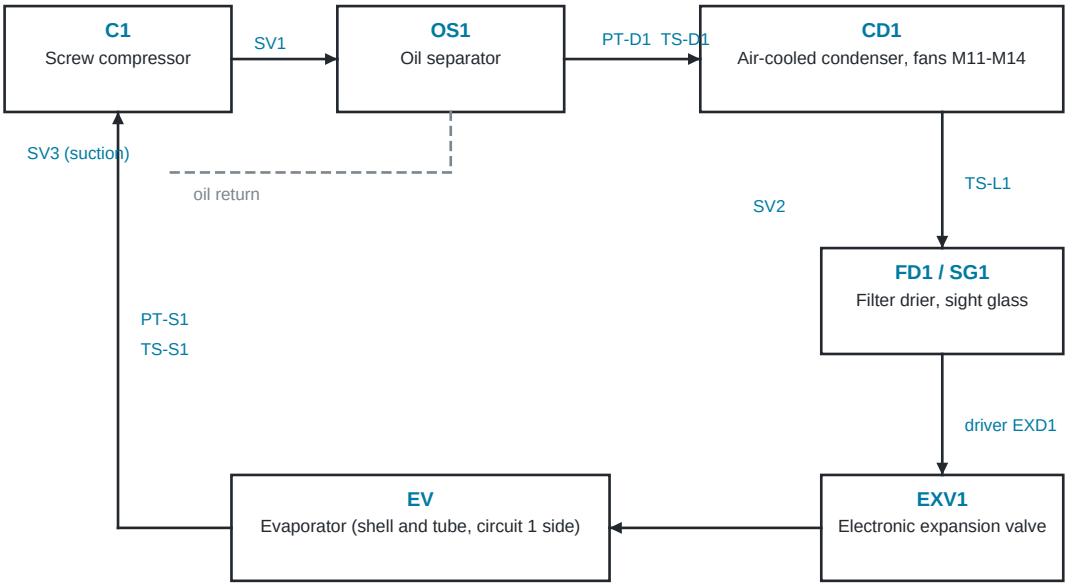

Figure 3-1 Refrigerant circuit 1. Circuit 2 is identical (tags C2, OS2, CD2 ...). Schematic, not to scale.

3.2 Description

Follow Figure 3-1 from the compressor. Hot discharge gas leaves compressor C1 through the discharge service valve SV1 and enters oil separator OS1. Separated oil returns to the compressor through the oil return line, driven by the pressure difference. The gas continues to condenser section CD1, where fans M11 to M14 reject heat to the outdoor air and the refrigerant condenses and is slightly subcooled.

Liquid leaves the condenser through the liquid line ball valve SV2, passes filter drier FD1 and sight glass SG1 with moisture indicator, and reaches the electronic expansion valve EXV1. The EXV is a stepper-motor valve driven by driver EXD1. It throttles the liquid to evaporating pressure and meters the flow into the circuit 1 side of the evaporator so that the suction gas leaves with the set superheat.

Suction gas returns to the compressor through the suction service valve SV3. Suction pressure transducer PT-S1 and suction temperature sensor TS-S1 sit on the suction line about 300 mm after the evaporator outlet. The controller calculates superheat from these two values, so TS-S1 must be strapped tight to the pipe and insulated. A loose or uninsulated TS-S1 reads warm and makes the EXV open further than it should.

3.3 Sensors and service connections

Tag	Function	Location
PT-S1 / PT-S2	Suction pressure transducer	Suction line, before SV3
TS-S1 / TS-S2	Suction temperature (superheat)	Suction line, 300 mm after evaporator outlet
PT-D1 / PT-D2	Discharge pressure transducer	Discharge line, after OS1
TS-D1 / TS-D2	Discharge temperature	Compressor discharge flange
TS-L1 / TS-L2	Liquid temperature (subcooling)	Liquid line, before FD1

Tag	Function	Location
HP1 / HP2	High pressure switch, manual reset	Discharge line
SV1, SV2, SV3	Service valves with 1/4 in. SAE access ports	Discharge, liquid, suction

4 Start-up and shutdown

4.1 Before starting

1 Power the controller at least 12 hours before the first start, and after any standstill with the main switch off. The crankcase heaters drive refrigerant out of the compressor oil. Starting with refrigerant in the oil foams the oil and can trip L-07.

2 Check that SV1, SV2 and SV3 on both circuits are fully open (back-seated) and capped.

3 Check that chilled water is flowing and the flow switch is made.

4 Check the alarm log for alarms that occurred while the unit was off.

4.2 Seasonal shutdown

Leave the controller powered and the service valves open. Do not pump the circuits down into the condenser for a seasonal shutdown: the NK-600 has no liquid receiver, and the charge does not fit in the condenser at high ambient temperature. Where the chilled water can freeze, drain the evaporator or use the glycol option.

5 Control logic

NordCtrl controls the leaving chilled water temperature to its setpoint by staging and loading the two compressors. The lead compressor rotates every 24 running hours. Each circuit is controlled on its own: its EXV controls suction superheat to the setpoint (default 6 K), and its fans control discharge pressure. At low ambient temperature the fans slow down to hold a minimum condensing pressure, which keeps enough pressure difference across the EXV.

When a circuit alarm occurs, only that circuit stops. Most circuit alarms restart automatically after a delay, but repeated alarms lock the circuit out. A locked-out circuit needs a manual reset at level 1 or higher. The restart rules for each alarm are given in Table 6-1.

Building management interface. The points below are the ones most often requested by site BMS integrators. The full point list, with BACnet object numbers, is in document NK-BMS-7 on the Service Portal. All Modbus points are holding registers, unit ID 1 by default, 19200 baud, 8E1.

Register	Point	Scaling	Access
40101	Leaving chilled water temperature	C x 10	R
40102	Entering chilled water temperature	C x 10	R
40110 / 40130	Suction pressure, circuit 1 / 2	bar a x 100	R
40111 / 40131	Discharge pressure, circuit 1 / 2	bar a x 100	R
40114 / 40134	Superheat, circuit 1 / 2	K x 10	R
40115 / 40135	EXV position, circuit 1 / 2	% x 10	R

Register	Point	Scaling	Access
40150	Active alarm, highest priority	L-code number, 0 = none	R
40200	Chilled water setpoint	C x 10	R/W
40201	Remote enable	0 = off, 1 = on	R/W
40202	Demand limit	% of capacity	R/W

Writing the setpoint from the BMS more often than once a minute wears the controller memory. Integrators should write only on change.

6 Alarms

6.1 Alarm handling

Active alarms appear on the display with the L-code, the circuit, and the time. The last 500 alarms and events are stored in Main > Alarms > Log, with the suction and discharge pressures, superheat, EXV position and water temperatures at the moment of the alarm. Export the log to USB as a CSV file (Main > Alarms > Export) and attach it to every warranty claim and every enquiry to Nordkyl support.

6.2 Alarm codes

Table 6-1. Auto: restarts after the delay shown. Lockout: manual reset required.

Code	Alarm	Trigger	Restart	Possible causes	First action
L-01	High discharge pressure	HP switch opens, or PT-D above 22.5 bar	Lockout	Dirty condenser coil, fan fault, air recirculation, overcharge	Check fans and coil. Reset HP switch by hand.
L-02	High discharge temperature	TS-D above 105 C for 30 s	Auto, 10 min; lockout after 3 in 24 h	High superheat, low charge, compressor wear	Check superheat and subcooling (6.4).
L-03	Low suction pressure	Suction pressure below 2.4 bar for 60 s	Auto, 5 min; lockout after 3 in 24 h	Low refrigerant charge (leak); iced or fouled evaporator, low water flow; blocked filter drier; EXV fault	Check EXV position and superheat in the EXV menu (7.4) before connecting gauges. See 6.3.
L-04	Freeze protection	Leaving water below 3.0 C	Lockout	Low water flow, setpoint too low, glycol too weak	Check flow and glycol before reset.
L-05	Compressor protection	Motor protection module trips	Lockout	Motor overheating, phase imbalance, winding fault	Measure winding resistance and supply.
L-06	Chilled water flow	Chilled water flow switch open for 10 s	Auto when flow returns	Pump stopped, strainer blocked, air in system	Check chilled water pump and strainer.
L-07	Oil level low	Optical level switch for 30 s	Auto, 10 min; lockout after 2 in 24 h	Oil foaming at start, poor oil return at low load	Check crankcase heater (4.1).
L-08	Phase fault	Phase loss or wrong sequence	Auto when corrected	Supply fault	Check supply.
L-09	Fan fault	EC fan reports fault	Auto	Fan motor, fan supply, Modbus link to fan	Read fan fault code in Service > Fans.

Code	Alarm	Trigger	Restart	Possible causes	First action
L-10	Sensor fault	Any sensor out of range	Lockout of affected circuit	Wire break, failed sensor	Check sensor and connector.
L-11	EXV driver fault	EXD reports open coil, short or low supply	Lockout	Coil or cable damage, driver failure	Check coil resistance and cable.
L-12	High superheat	Superheat above 15 K for 10 min with EXV at 100 %	Warning only	Low charge, restriction before the EXV, EXV not opening	Treat as early warning of L-03. See 6.3.
L-13	Low superheat	Superheat below 1 K for 3 min	Auto, 5 min	TS-S1 loose or uninsulated, EXV not closing	Check TS-S mounting, then EXV (7.4).
L-14	Communication loss	No response from EXD or fan bus for 30 s	Auto	Bus cable, termination	Check bus wiring and termination.

6.3 Diagnosing L-03

L-03 is the most frequent alarm on air-cooled chillers and the one most often misdiagnosed. Low suction pressure means the evaporator is getting too little refrigerant, or is taking up too little heat. A low charge is only one of the causes. An EXV that does not open as far as the controller commands starves the evaporator in exactly the same way, and produces the same alarm.

Before connecting gauges, open the EXV menu for the alarmed circuit and the alarm log entry. Then compare:

Reading	Low charge (leak)	EXV restricted or not opening	Blocked filter drier	Iced evaporator / low flow
Superheat	High	High	High	Low or normal
EXV position shown	At or near 100 %	Any value; often normal	At or near 100 %	Low
Subcooling	Low (below 2 K)	Normal or high	Normal before the drier	Normal
Sight glass	Bubbles	Clear	Bubbles after the drier	Clear
Temperature drop across FD1	Below 1 K	Below 1 K	Above 3 K	Below 1 K

The EXV position shown on the display is calculated from the number of steps the driver has sent. It is not measured at the valve. A valve that is sticking reports a normal position while passing far less refrigerant than the position suggests. Use the manual test in 7.4.3 to check whether the valve actually responds.

> **IMPORTANT**
> Do not add refrigerant to clear L-03. If the charge is low, there is a leak: find and repair it first (chapter 8). Adding refrigerant to a circuit with a restricted EXV floods the condenser, raises the discharge pressure, and in warm weather leads to L-01.

6.4 Typical operating values

At nominal conditions (chilled water 12/7 C, ambient 35 C, full load). Values at part load and low ambient are lower for discharge pressure and EXV position.

Value	Typical range
Suction pressure	3.4 - 3.9 bar a
Suction superheat	5 - 8 K (setpoint 6 K)

Value	Typical range
Discharge pressure	10.5 - 14.0 bar a
Discharge temperature	60 - 80 C
Liquid subcooling	4 - 7 K
EXV position, stable load	15 - 85 %
Temperature drop across filter drier	below 1 K

7 Service menus

7.1 Access

Log in at level 2 (Main > Login). The Service menu times out to level 1 after 30 minutes without input. Any manual override still active at time-out is cancelled and the circuit returns to automatic control.

7.2 Circuit menu

Main > Service > Circuits > Circuit 1 (or 2) shows all live values of the circuit on one page: suction and discharge pressure and saturation temperatures, superheat, subcooling, compressor load, fan speed, EXV position and the run hours and starts of the compressor. Trend graphs of the last 24 hours are under the graph symbol.

7.3 Filter drier check

Measure the liquid line temperature on both sides of the filter drier with contact thermometers while the circuit runs at full load. A drop of more than 3 K indicates a restricted core. Replace the core (63-0410). A frosted or cold drier shell is a sure sign of restriction.

7.4 EXV service menu

Main > Service > Circuits > Circuit 1 (or 2) > EXV. Available from controller software 7.0. On units with software 6.9, the EXV position is shown in the circuit menu only and the manual test is not available; update the software first (Service Portal, file NC7-SW). The valve type fitted is set under Main > Service > Configuration > EXV type and must read EXV-2.

7.4.1 Display fields

Field	Meaning
EXV request	Opening requested by the superheat controller (or set by hand in MAN), 0 - 100 %
EXV position	Opening the driver has stepped the valve to, 0 - 100 %. Calculated from steps sent, not measured.
Steps	Driver step count, 0 - 2,500 (full stroke of the EXV-2)
SH act / SH set	Actual superheat / superheat setpoint (default 6.0 K)
Ps / Tsat / Tsuc	Suction pressure / saturation temperature / suction temperature from TS-S
Mode	AUTO, MAN (manual override), INIT (re-synchronising)
Drv	Driver status: OK, OPEN COIL, SHORT, LOW SUPPLY, OVERTEMP
Sync	Hours since the last re-synchronisation

7.4.2 Normal behaviour. In stable operation the EXV position should modulate between 15 and 85 %. The position follows the load: lower at part load and low ambient, higher at full load and high ambient. Small continuous movement is normal. Look for these patterns:

- Position at 100 % for more than 10 minutes with superheat above 12 K: the evaporator is starved. Low charge, a restriction before the valve, or a valve that is not opening. See 6.3.
- Position below 15 % at full load with low superheat: check the mounting and insulation of TS-S before the valve.
- EXV position changing on the display while suction pressure and superheat do not respond: the driver is stepping but the valve is not moving. The driver cannot detect this. Carry out the manual test.
- EXV position lagging EXV request by more than a few seconds: driver fault or low driver supply. Check Drv.
- Position hunting between extremes every few minutes: check the superheat setpoint and the TS-S mounting.

7.4.3 Manual test. Carry out with the circuit running at a stable load of at least 50 %.

1 Note EXV request, EXV position, SH act and Ps in AUTO.

2 Set Mode to MAN. The valve holds its present position.

3 Set EXV request to 30 % and wait 2 minutes. EXV position follows within a few seconds. Note Ps and SH act.

4 Set EXV request to 70 % and wait 2 minutes. Suction pressure should rise by at least 0.4 bar and superheat should fall. Hold a screwdriver or stethoscope to the valve body during the change: the stepping of the motor is audible.

5 Set Mode to AUTO. MAN also ends automatically after 15 minutes.

6 If suction pressure and superheat did not respond, or the motor could not be heard, the valve is faulty. Check the coil resistance (45 - 55 ohm per phase) and the cable to rule out the driver, then replace the EXV assembly (7.5).

7.4.4 Re-synchronisation. INIT drives the valve fully closed with 10 % extra steps to re-zero the step count. The controller does this automatically at every start and every 24 hours. Start it by hand (Service > Circuits > Circuit n > EXV > INIT) after replacing a driver or valve. The circuit is held off for about 40 seconds.

7.5 Replacing the EXV assembly

1 Isolate the unit. Close SV2 and SV3 of the circuit and recover the refrigerant from the section between them, or from the whole circuit (8.4).

2 Disconnect the EXV cable at the valve (EXV-2 rev 2, 63-1170: grey cable, 4-pin connector). Unbrazing is not needed: the assembly is fitted with flanged connections and two O-rings.

3 Fit the new assembly with new O-rings, lubricated with POE oil. Tighten the flange bolts evenly.

4 Replace the filter drier core (63-0410). The circuit has been opened.

5 Pressure test and evacuate in accordance with EN 378-2 and local regulations (8.4). Do not exceed the low side design pressure.

6 Open SV2 and SV3, charge by weight if refrigerant was removed, and start the circuit. Run INIT, then the manual test (7.4.3). Record the work in the service record (Annex A).

8 Refrigerant handling and F-gas obligations

8.1 R-513A

R-513A is an azeotropic mixture of R-1234yf (56 %) and R-134a (44 %), GWP 631, safety group A1. It behaves as a single substance, but Nordkyl recommends charging as liquid. Use POE oil only. Do not mix with other refrigerants. Recovered R-513A must go to an approved reclaimer, not back into another unit, unless it has been recovered into a clean dedicated cylinder from a circuit with no compressor burn-out.

8.2 Legal obligations in the EU

Under Regulation (EU) 2024/573 the operator of the unit is responsible for the following. The service partner carries out the work and supplies the records.

- Leak checks: the NK-600 holds 30.3 t CO2 equivalent, so it must be checked for leaks at least every 12 months, or every 24 months if a leak detection system is fitted (option NK-LD).
- Repair: a leak must be repaired without undue delay. The repair must be checked by a certified person within one month to confirm it is effective.
- Records: for at least five years, the quantity and type of refrigerant in the unit, every quantity added or recovered, the result of every leak check, and the name and certificate number of the person who did the work.
- Personnel: only certified persons may check for leaks, recover, install, service or repair the circuits.

Outside the EU, national rules apply (for example EPA Section 608 in the United States). Where national rules are less strict than this manual, follow this manual.

8.3 Low charge

Refrigerant does not wear out or get used up. A circuit that is short of charge has a leak. Adding refrigerant without finding and repairing the leak is not an approved repair: it releases more refrigerant, breaks the repair obligation above, and is not covered by the Nordkyl warranty. Before deciding that the charge is low, rule out the other causes of low suction pressure (6.3).

Check for leaks with a calibrated electronic detector, below the joints, because R-513A is heavier than air. Typical leak points on the NK-600: flare connections at the transducers, Schrader cores, the oil return line fittings (vibration), and the condenser header joints.

8.4 Recovery, repair and charging

1. Recover the refrigerant into a dedicated, labelled recovery cylinder. Weigh and record the quantity recovered.
2. Repair the leak. Replace the filter drier core 63-0410 whenever the circuit has been opened.
3. Strength and tightness test with dry nitrogen in accordance with EN 378-2 and local regulations. Never exceed the low side design pressure, and never use oxygen or compressed air.
4. Evacuate to the level required by local regulations and confirm with a standing vacuum test before charging.
5. Charge by weight as liquid through the liquid line access port: 24.0 kg per circuit. After 30 minutes at full load, check subcooling and the sight glass against 6.4.
6. Record the quantities, the leak location and the repair in the service record and the operator's F-gas log.

9 Planned maintenance

The intervals below apply to normal duty. Halve the coil cleaning interval near roads, farmland, airports and the sea.

Task	Interval	Ref.
Read alarm log; investigate any alarm that occurred more than twice	Monthly	6.1
Look for oil traces at joints, valves and transducers	Monthly	8.3
Clean condenser coils	6 months	9.1
Statutory leak check, both circuits	12 months (24 with NK-LD)	8.2
EXV manual test, both circuits	12 months	7.4.3
Filter drier temperature drop	12 months	7.3
Crankcase heaters: measure current	12 months	4.1
Oil sample: acidity and moisture	12 months	-
Test high pressure switches HP1 / HP2	12 months	6.2
Test flow switch and freeze protection	12 months	6.2
Retighten power terminals to the values in the wiring diagram	12 months	-
Check transducers against a reference gauge	36 months	3.3
Replace safety valves	5 years or after discharge	1.3
Compressor inspection (bearings, slide valve)	40,000 run hours	-
Replace filter drier core 63-0410	When opened, or drop above 3 K	7.3

9.1 Condenser coil cleaning

Isolate the unit. Clean from the inside of the V-coil outwards with low-pressure water (below 5 bar) or a coil cleaner approved for aluminium fins. Never use a pressure washer at close range: bent fins block the airflow as effectively as dirt. Straighten damaged fins with a fin comb.

9.2 Water side

Clean the chilled water strainer at the evaporator inlet every 12 months or when the water pressure drop rises 15 % above the commissioning value. Check the glycol concentration every 12 months on units with the glycol option.

10 Spare parts

Order spare parts through the Nordkyl Service Portal with the unit serial number. Most components are selected by serial number (compressor, fans, controller, sensors) and are not listed here. The parts below are common to all NK-600 units covered by this manual and are recommended as van stock for service partners.

Part no.	Description	Qty per unit	Notes
63-0410	Filter drier core, NK series	2	Replace whenever a circuit is opened
63-1170	EXV-2 expansion valve assembly, rev 2, NK-600, with O-rings	2	One per circuit. Grey cable, 4-pin connector

Use only Nordkyl EXV assemblies. EXV assemblies from other makers have different step counts and will not be recognised by the driver.

11 Warranty

Nordkyl warrants the NK-600 against defects in material and workmanship for 24 months from commissioning or 30 months from delivery, whichever ends first. The compressors are covered for 36 months from commissioning if the commissioning report was registered on the Service Portal within 30 days.

The warranty requires that planned maintenance has been carried out as described in chapter 9 by an authorised service partner, and that records are available. It does not cover refrigerant, oil, filter drier cores, damage from frozen evaporators, operation outside the limits in 2.2, or work by uncertified persons.

Submit claims through the Service Portal with the serial number, the alarm log export (6.1), the service record and, for parts, the removed part. Nordkyl Product Support answers claims within 10 working days.

Nordkyl A/S, Product Support, Hedeager 21, 8200 Aarhus N, Denmark. Telephone +45 8730 4410. support@nordkyl.dk

A Service record

Date	Circuit	Work done / findings	Refrigerant added / recovered (kg)	Technician, cert. no.

B Leak check record (Regulation (EU) 2024/573)

Keep this record with the unit or in the operator's electronic F-gas register for at least five years. One line per circuit per check. Where a leak was found, enter the repair and the date of the follow-up check, which must take place within one month of the repair.

Item	Entry
Unit serial number	
Refrigerant / charge per circuit	R-513A / 24.0 kg
CO2 equivalent per unit	30.3 t
Leak detection system fitted (NK-LD)	Yes / No
Required check interval	12 months / 24 months with NK-LD

Date	Circuit	Method	Result / leak location	Repair date	Follow-up check	Name, cert. no.